Formal Language-Based

Universidad Icesi

Computationand Discrete Structures III

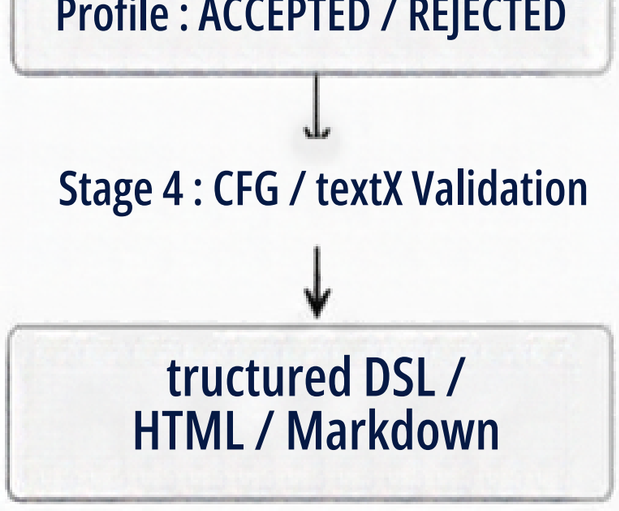
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SECTION 1 : ABSTRACT & PROBLEM STATEMENT

Problem: Resumes are unstructured and vary widely in syntax, naming conventions, and missing information.

Proposed Solution: Explainable, deterministic screening architecture" por "an inspectable, rule-based screening prototype

SECTION 2 : SYSTEM ARCHITECTURE & PIPELINE



SECTION 3 : FORMAL MODELS DEFINITION

Stage 1 : Information Extraction

Identify tokens with Python patterns: Languages: JavaScript|JS|Python|Java|TypeScript
ML Frameworks: TensorFlow|PyTorch|Scikit-learn|sklearn
Databases:PostgreSQL|Postgres|MySQL|Mongo DB|SQL|Pandas

Stage 2 : Qualification Normalization (FST)

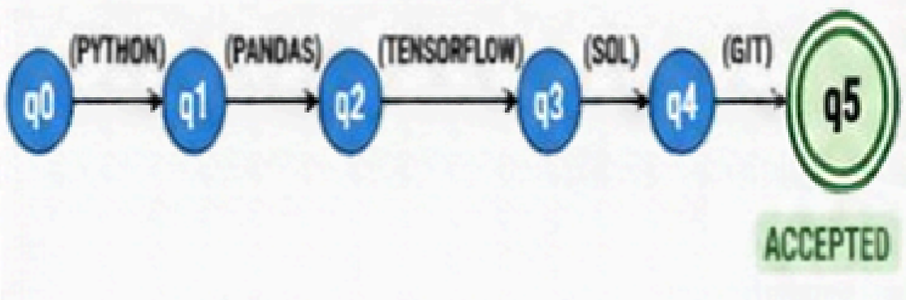
Formal definition : $M_{FST} = (Q, \Sigma, \delta, \omega, q_0, F)$
Normalizes token variations:

- JS → JAVASCRIPT
- sklearn → SCIKIT_LEARN
- postgres → POSTGRESQL

Canonical sorting preps ordering: Language → Framework → Database → API → Infrastructure → Git (order depends on profile)

Stage 3 : Pattern Recognition (DFA)

Formal definition : $M_{DFA} = (Q, \Sigma, \delta, q_0, F)$
Linear state - transition Diagram :



Stage 4: Candidate Profile Language

Defines structured candidate records including personal details, normalized skills, and classification results. Output: validated DSL, plus HTML and Markdown reports

SECTION 4: EXPERIMENTAL RESULTS & VERIFICATION

100 Automated Test
100%



CaseStudy

Python, Pandas, NumPy, Scikit-learn, TensorFlow, SQL, Git → Normalization & DFA → **ACCEPTED** (Machine Learning Engineer)

Case Study (Incomplete)

Missing critical credentials → **REJECTED** (Incomplete Path)

SECTION 5 : CONCLUSION & KEY TAKEAWAYS

Rule-based and inspectable: The same input always produces the same output. Tests use synthetic data and do not measure real-world accuracy or bias.
Modular Architecture: Unified multi-stage pipeline processing heterogeneous input texts into canonical formats and structured DSL specifications.
Robustness: Verified with 100 automated tests, including end-to-end runs on 6 synthetic resumes.